

Application Note:

CJM410 Uart to WiFi with TCP/UDP Connection Demo

Rev 1.4

1. EVB Hardware Setup

The CJM410 EVB supports two UART interfaces : Debug UART and High Speed UART. The definition of header pins is as follows:

Interface	Signal	Header pins
Debug UART	RXD	J1.6
	GND	J1.7
	TXD	J1.8
High Speed UART	RXD	JP6.4
	GND	JP6.5
	TXD	JP6.6
	RTS	JP6.8
	CTS	JP6.10

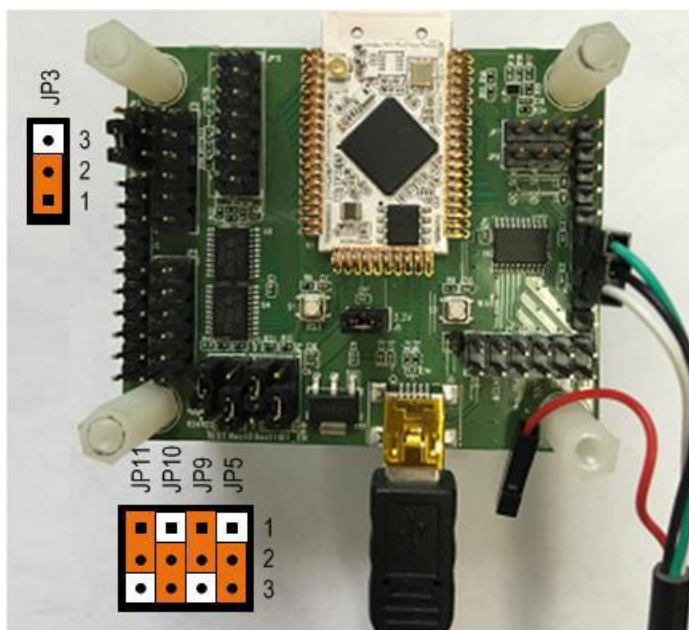


Figure 1

As the Figure 1, the Debug UART header pins are connected through USB to UART adapter cables to serial port on a Windows PC, and the USB cable is connected to Windows PC to power up the WiFi module.

2. Environment Setup

The CJM410 WiFi module supports AT commands via console and Web interface to configure the wireless and network settings. Please refer to the *CJM410 WiFi module AT commands.pdf* document for detail description.

The following demo applications are the setting examples using AT commands or Web interface to configure CJM410 as AP mode or client mode, and to create TCP/UDP connection for data transmission between wireless network and UART port.

2.1 Demo applications

The Figure 2 is to configure the CJM410 as wireless AP mode, and running TCP/UDP server on it. The Figure 3 is to configure the CJM410 as wireless client mode, and running TCP/UDP client on it.

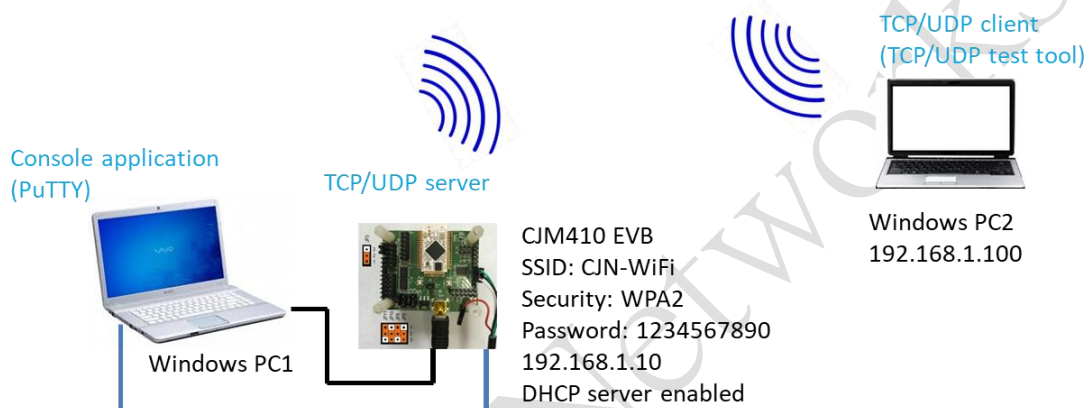


Figure 2



Figure 3

2.2 Run the console application (ex. Putty) on Windows PC1 with port setting:

115200 bps, 8, N, 1.

2.3 Run TCP or UDP test tool on Windows PC2.

3. CJM410 Configuration

3.1 Configuration via CLI command

Please type the following commands on Windows PC1 console application.

3.1.1 Switch UART port to command mode

When entering into command mode, the “shell>” prompt will appear.

Step	Command
1. Enter command mode	+++ shell>

3.1.2 Wireless settings

- The CJM410 is configured as wireless AP mode. (see Figure 2)
The default wireless settings are as Figure 2: The operating mode is AP mode, the SSID is CJN-WiFi and the security mode is WPA2 with CCMP (AES) encryption, password is 1234567890. If you don't want to change the settings, you can ignore the following steps.

Step	Command
1. Set operating mode	AT+WOP=1 +ok
2. Set SSID and channel	AT+WAP=CJN-WiFi,6 +ok
3. Set security mode	AT+WAPSEC=WPA2 +ok
4. Set WPA2 settings	AT+WAPWPA=CCMP,1234567890 +ok

- The CJM410 is configured as wireless client mode. (see Figure 3)

Step	Command
1. Set operating mode	AT+WOP=0 +ok
2. Set SSID	AT+WSTA=CJN-WiFi +ok
3. Set security mode	AT+WSTASEC=WPA2 +ok
4. Set WPA2 settings	AT+WSTAWPA=CCMP,1234567890 +ok

3.1.3 IP Settings

If the CJM410 is configured as wireless AP mode, the CJM410 default is fixed IP 192.168.1.10, and the DHCP server will be enabled automatically. The IP range is between 192.168.1.100 to 192.168.1.200 for DHCP clients. If you change the IP domain, the DHCP IP domain will also be changed the same as IP setting. If the CJM410 is configured as wireless client mode, the DHCP client default is enabled to get the IP address from DHCP server.

● Set Fixed IP address

Step	Command
1. Set IP address	AT+NIPSET=192.168.1.10,255.255.255.0 +ok

3.1.4 Configure CJM410 to be TCP Server

This demo application is to transfer data between Windows PC1 and Windows PC2 by TCP protocol. The CJM410 is running TCP server, and Windows PC2 is running the TCP client to connect to CJM410. Assume the TCP server port number is 5000.

Step	Command
1. Set TCP protocol	AT+NSOCK=TCP +ok
2. Set TCP server port number	AT+NTCPS=5000 +ok

3.1.5 Configure CJM410 to be TCP client

This demo application is to transfer data between Windows PC1 and Windows PC2 by TCP protocol. Windows PC2 is running the TCP server, and the CJM410 is running TCP client to connect to Windows PC2. Assume the TCP server port number is 5000.

Step	Command
1. Set TCP protocol	AT+NSOCK=TCP +ok
2. Set TCP server IP address and port number	AT+NTCPC=192.168.1.100, 5000 +ok

3.1.6 Configure CJM410 to be UDP server

This demo application is to transfer data from Windows PC1 to Windows PC2 or PC2 to PC1 by UDP protocol. The CJM410 is running UDP server, and Windows PC2 is running the UDP client. Assume the UDP server port number is 5000.

Step	Command
1. Set UDP protocol	AT+NSOCK=UDP +ok
2. Set UDP server port number	AT+NUDPS=5000 +ok

3.1.7 Configure CJM410 to be UDP client

This demo application is to transfer data from Windows PC1 to Windows PC2 or PC2 to PC1 by UDP protocol. The CJM410 is running UDP client, and Windows PC2 is running the UDP server. Assume the UDP server port number is 5000.

Step	Command
3. Set UDP protocol	AT+NSOCK=UDP +ok
4. Set UDP server IP address and port number	AT+NUDPC=192.168.1.100,5000 +ok

3.1.8 Apply the configuration

After the configuration is finished, please save all the settings to flash, and reboot the CJM410 module to apply the configuration.

Step	Command
1. Save all the settings to flash	AT+SAVE= +ok
2. Reboot the module	AT+RST=

Or

Step	Command
1. Save and apply the configuration	AT+APPLY=

3.1.9 Query the status

Assume the CJM410 is running as wireless client mode and TCP client, and you can query the current status of CJM410 module by the following commands:

Step	Command
1. Enter command mode	+++ shell>
2. Query wireless link status	AT+WLINK= +ok=ssid: CJN-WiFi, channel: 6, rssi: 30 dB
3. Query the system IP address	AT+NIPSET=? +ok=IP: 192.168.1.10, Mask: 255.255.255.0, Gateway: 192.168.1.254
4. Query TCP client connection status	AT+NTCPLINK= +ok=connected
5. Return to data mode	ATO=

3.2 Configuration via web interface

The CJM410 supports simple web UI to configure wireless and network settings. You can connect to the CJM410 by WiFi. The CJM410 WiFi default is AP mode, SSID is "CJN-WiFi, security is WPA2 with CCMP (AES) encryption, and password is "1234567890".

3.2.1 Logging into the Web interface

You can log into the Web interface using a computer via a standard Web browser. In the address or location bar, type the IP address of CJM410. The default IP address is 192.168.1.10.

3.2.2 Navigating the Web interface

You can select the operation mode to be AP mode or client (STA) mode. The AP mode settings are as Figure 4 to configure the demo application as Figure 2, and if the client (STA) mode is selected, the Web interface will appear as Figure 5 to configure the demo application as Figure 3.

192.168.1.10

WiFi Configuration

Operation mode : ☒ AP ☐ STA

SSID : CJN-WiFi

Channel : 6

Password : (8 ~ 63 characters)

Network

Protocol : ☒ TCP ☐ UDP

Port : 5000

Submit Reset System

Current firmware version : 2.1.2

Firmware Update : 選擇檔案 未選擇任何檔案

Flash image...

Figure 4.

192.168.1.10

WiFi Configuration

Operation mode : ☐ AP ☒ STA

SSID : CJN-WiFi

Security : ☐ Open network

Password : (8 ~ 63 characters)

Network

Protocol : ☒ TCP ☐ UDP

Port : 5000

Remote IP : 192.168.1.20

Submit Reset System

Current firmware version : 2.1.2

Firmware Update : 選擇檔案 未選擇任何檔案

Flash image...

Figure 5.

The settings used for AP mode are listed as below

Element	Description
Wireless	
SSID	Specify SSID between 1 ~ 32 characters
Channel	Select the channel used by wireless network. FCC: 1 ~ 11, CE: 1 ~ 13
Password	Set the password for AP mode security. The security of AP mode default is WPA2 with CCMP (AES) encryption. Please specify 8 ~ 63 characters.
Network	
Protocol	Set TCP or UDP server
Port	Set server port number

The settings used for client mode are listed as below

Element	Description
Wireless	
SSID	Specify connecting AP SSID between 1 ~ 32 characters
Security	Set security for client mode. The open network is no security. If this checkbox is unchecked, the security is set to be WPA2 with encryption CCMP (AES).
Password	If the checkbox of open network is unchecked, the password field is enabled to input for client mode security. Please specify 8 ~ 63 characters.
Network	
Protocol	Set TCP or UDP client
Port	Set remote server port number
Remote IP	Set the remote server IP address

3.2.3 Apply the configuration

After the configuration is finished, please press the "Submit" button to save the settings, and press the "Reset System" button to reboot the system to apply the configuration.

Note:

1. There are many console applications for Windows. For example, TeraTerm, RealTerm, and PuTTY ...etc. are freewares, and you can download them from internet. Please refer to the following web sites:

<https://tssh2.osdn.jp/>

<http://realterm.sourceforge.net/>

<http://www.chiark.greenend.org.uk/~sgtatham/putty/download.html>

2. TCP test tool and UDP test tool are freeware, and you can download them from the following web site:

<https://www.simplecomtools.com/productcart/pc/viewCategories.asp?idCategory=5>

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